

PORTLAND INTL JETPORT, ME, USA

WMO: 726060

Lat: 43.650N

Lon: 70.317W

Elev: 14

StdP: 101.16

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14764

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.7	-15.0	-25.5	0.4	-16.1	-22.9	0.5	-13.5	12.1	-2.6	10.9	-4.0	3.1	280	0.552

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.5	30.3	22.0	28.5	21.0	26.9	20.1	23.4	28.4	22.4	26.7	21.4	25.1	4.5	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.7	16.4	26.0	20.9	15.6	24.6	20.1	14.8	23.7	69.6	28.5	65.8	26.8	62.4	25.1	27.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
10.4	8.7	7.7		-22.4	33.6	3.1	1.7	-24.7	34.8	-26.5	35.9	-28.2	36.8	-30.5	38.1	
				-22.8	25.1	3.0	1.0	-25.0	25.8	-26.8	26.4	-28.5	27.0	-30.7	27.7	
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Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.5	-4.7	-3.4	0.9	6.9	12.4	17.6	21.0	20.4	16.3	10.1	4.3	-1.0
	DBStd	9.82	5.83	5.25	4.78	3.86	3.68	3.65	2.79	2.71	3.66	3.98	4.50	4.96
	HDD10.0	1808	457	375	283	108	14	0	0	0	1	48	178	342
	HDD18.3	3828	715	608	539	344	190	57	7	11	81	256	422	599
	CDD10.0	1244	0	0	2	14	88	227	341	321	192	52	7	1
	CDD18.3	223	0	0	0	0	5	34	89	73	21	1	0	0
	CDH23.3	1603	0	0	2	8	69	273	635	473	133	10	0	0
	CDH26.7	405	0	0	1	2	19	73	168	109	32	1	0	0
Wind	WSAvg	3.4	3.5	3.7	4.1	4.0	3.5	3.1	2.9	2.8	2.9	3.3	3.3	3.4
Precipitation	PrecAvg	1140	91	85	97	104	92	85	78	76	86	110	123	112
	PrecMax	1687	303	180	236	254	233	229	197	387	223	375	343	255
	PrecMin	737	7	1	21	7	13	19	16	12	16	16	24	30
	PrecStd	215	54	41	45	50	49	43	39	58	40	81	63	59
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.0	11.0	18.9	23.9	29.4	31.6	33.1	32.0	30.4	24.8	17.8	13.6
		MCWB	9.7	7.3	12.0	14.7	18.8	22.4	23.4	23.1	22.4	17.5	13.0	10.0
	2%	DB	8.4	8.0	12.4	18.5	24.9	28.8	30.2	29.4	26.7	20.8	15.1	10.4
		MCWB	6.4	5.2	7.4	10.9	16.9	20.8	22.0	21.9	20.0	15.6	12.1	8.3
	5%	DB	5.8	6.0	9.5	15.7	21.8	26.4	28.4	27.6	24.4	18.3	13.3	8.5
		MCWB	3.9	3.5	5.8	9.5	15.2	19.1	21.4	20.9	18.8	14.5	10.8	6.6
	10%	DB	3.4	4.1	7.4	13.3	19.1	24.2	26.8	26.0	22.4	16.4	11.5	6.3
		MCWB	1.4	2.0	4.3	8.2	13.6	18.2	20.5	19.9	17.9	13.2	9.2	4.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.2	8.4	12.7	15.5	20.4	23.9	25.0	24.5	23.0	19.2	15.0	11.3
		MCDB	10.6	9.9	16.9	21.7	26.2	29.9	30.9	29.8	28.4	21.9	16.8	12.6
	2%	WB	6.8	5.9	8.9	12.8	18.3	22.1	23.5	23.2	21.5	16.9	13.1	8.9
		MCDB	7.8	7.6	11.6	16.5	23.0	26.6	28.4	27.7	25.0	19.4	14.2	10.0
	5%	WB	4.0	3.9	6.5	10.6	16.3	20.7	22.5	22.1	20.2	15.4	11.3	6.8
		MCDB	5.6	5.4	8.9	13.9	20.9	24.6	26.8	25.8	22.9	17.7	12.6	8.1
	10%	WB	1.6	2.2	4.6	9.0	14.4	19.3	21.5	21.1	18.9	14.0	9.4	4.4
		MCDB	3.1	4.0	6.8	12.7	18.2	23.0	25.3	24.2	21.4	16.0	11.3	6.1
Mean Daily Temperature Range	5% DB	MDBR	8.7	9.2	8.5	9.6	9.5	9.5	9.5	9.8	9.8	9.6	8.6	8.3
		MCDBR	10.1	10.1	11.3	13.7	14.5	12.8	11.7	11.8	12.1	12.0	10.2	10.5
		MCWBR	9.2	8.2	8.0	8.3	7.9	6.6	5.5	5.9	6.8	7.8	8.1	9.1
	5% WB	MCDBR	10.0	9.6	10.4	11.6	12.8	11.3	10.1	10.1	9.9	10.0	8.7	10.1
		MCWBR	9.6	8.5	8.1	8.3	7.8	6.7	5.4	5.7	6.4	7.6	7.9	9.6
Clear-Sky Solar Irradiance	taub	0.291	0.304	0.329	0.356	0.390	0.442	0.456	0.423	0.379	0.338	0.319	0.295	
	taud	2.451	2.421	2.434	2.416	2.366	2.241	2.216	2.299	2.400	2.510	2.526	2.514	
	Ebn at Noon	845	895	912	910	883	833	816	833	846	845	807	809	
	Edh at Noon	73	89	100	110	120	136	138	123	103	81	68	63	
All-Sky Solar Radiation	RadAvg	1.67	2.55	3.68	4.61	5.16	5.54	5.82	5.29	4.26	2.74	1.78	1.33	
	RadStd	0.12	0.19	0.23	0.49	0.59	0.46	0.40	0.27	0.24	0.25	0.16	0.14	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.39	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+85	+30
(47)	Regional (55 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+77	N/A

Nomenclature: See separate page